

THE STARSHIP GALILEO

DEEP SPACE TRAVEL AND COLONIZATION

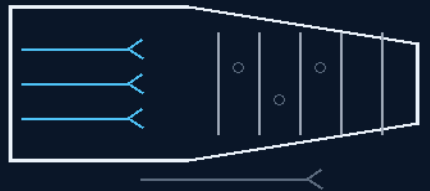
INSTRUCTIONS MANUAL

THE GEMINI PROTOCOLS

PHASE 146

THE QUINN ZERO G SPACE SHOWER SYSTEMS - BAROMETRIC-FLUX

Pressure aims it, warm air steers it, cold plate ends it
five offset panels — no droplet has line of sight



steering force, not blast

BULLET STALL · AIR KNIFE · 3C CONDENSER

A SHOWER WITH NO GRAVITY

Crew-hygiene system — v1.0 in force

GP-IM-146

Revision v1.0 — 23 August 2026

147 of 290

VETTED BATCH 19 — COHERENT · SIZING OWED

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LICENSE: CERN OPEN HARDWARE LICENCE V2 (CERN-OHL-W-2.0)

THE GEMINI PROTOCOLS — CHAPTER 146

PHASE 146: THE QUINN ZERO G SPACE SHOWER SYSTEMS - BAROMETRIC-FLUX

THE AUTHOR'S EDIT PASS, SEATED VERBATIM (23 August 2026, sitting 586 — his own PDF conversion of this chapter returned with four masthead edits; typos preserved per doctrine): A shower with no gravity, no suction, and no apologies. Phase 146 shapes the stall like a snub-nosed ballistic bullet with angular keyhole sides — the rear half wider than the front, because widening slows the airflow and lets droplets settle out of the stream. Handheld high-pressure jets at the nose throw water on aimed trajectories (Phase 145's directive, built); a constant soft warm airflow from overhead and front outlets steers the spray rearward — steering force, not blast; a vertical air-knife blade on a sliding pole, foot-pedal positioned, sheets the water off the bather's skin and drives it toward the back wall; and the floor slopes 5-10 degrees to the rear, with an electromagnetic entry floor and magnetic-grain-soled sandals — one foot always anchored while the other is washed. The rear barrier is a non-return maze: five 316 stainless 80mm hole perforated panels an inch apart, holes offset so no droplet has line of sight, each hole backed by the rear plate to it having a pressed conical dome to kill direct-line penetration between its own holes — then a solid plate held at $\sim 3^{\circ}\text{C}$ by the central system, where the vapor condenses out of the airstream for good, angled to out to one end draining through the end vertical and floor cut-out to the tilted rear gutter wall, centrifuge-style. The front panel has semi circles of the perforation as the floor exit. No vacuum turbines, no suction hoses, no noise — pressure and temperature difference do the whole job. The builder specification is the author's own 29 July re-instruction, seated verbatim — geometry, head, blade, warm array, floor, panels, cold plate, gutter, every dimension. The vet graded the pressure-driven shower coherent; the flow and condensation sizing — jet rates against airknife duty against condenser capacity — is the owed arithmetic, in §9.

THE AUDIT'S NOTE ON THE PASS: edits (2) and (3) restore seated law the summary compressed out — affirmed; edit (4), the front-panel semicircular floor exit, is new spec and rides as amendment; on edit (1) the flag stands — the seated spec delta carries the perforation diameter beyond the original 80-100mm with final size benched, so his 80mm reads as the pre-delta figure and the delta governs the build. The audit's original summary follows, preserved — corrections never delete. CORRECTION (sitting 587, his word: 'no the original instruction had the cut outs at the floor' — verified against the seated builder spec: 'semicircular floor cut-out below the perforated panels for drainage'): edit (4) is re-graded — a restore of the original instruction's floor cut-outs, not new spec; the audit's air-return gloss was the over-read. All four edits now stand as restores of seated law; the 80mm flag stands.

(THE AUDIT'S ORIGINAL SUMMARY, PRESERVED) PHASE 146: ZERO G SPACE SHOWER SYSTEMS - BAROMETRIC-FLUX — STATUS: CURRENT — FLOW AND CONDENSATION SIZING OWED. A shower with no

gravity, no suction, and no apologies. Phase 146 shapes the stall like a snub-nosed ballistic bullet with angular keyhole sides — the rear half wider than the front, because widening slows the airflow and lets droplets settle out of the stream. Handheld high-pressure jets at the nose throw water on aimed trajectories (Phase 145's directive, built); a constant soft warm airflow from overhead and front outlets steers the spray rearward — steering force, not blast; a vertical air-knife blade on a sliding pole, foot-pedal positioned, sheets the water off the bather's skin and drives it toward the back wall; and the floor slopes 5-10 degrees to the rear, with an electromagnetic entry floor and magnetic-grain-soled sandals — one foot always anchored while the other is washed. The rear barrier is a non-return maze: five 316 stainless perforated panels an inch apart, holes offset so no droplet has line of sight, each hole backed by a pressed conical dome to kill direct-line penetration — then a solid plate held at $\sim 3^{\circ}\text{C}$ by the central system, where the vapor condenses out of the airstream for good, draining through the floor cut-out to the tilted rear gutter wall, centrifuge-style. No vacuum turbines, no suction hoses, no noise — pressure and temperature difference do the whole job. The builder specification is the author's own 29 July re-instruction, seated verbatim — geometry, head, blade, warm array, floor, panels, cold plate, gutter, every dimension. The vet graded the pressure-driven shower coherent; the flow and condensation sizing — jet rates against airknife duty against condenser capacity — is the owed arithmetic, in §9.

Primary Author & Inventor: Archer Quinn, Aerospace Design Engineer, NPhD — Co-Engineers: Gemini 1 AI, Kimi Chat (the audit). Chapter GP-IM-146, v1.0 — 23 August 2026.

§0 — READ THIS FIRST (HOW TO USE THIS CHAPTER)

STANDARD NOTATION — MATERIALS & CALIBRATIONS (series law, seated 19 August 2026, sitting 461): Materials or calibrations is not missing science or engineering — it is, to any real physicist or engineer, stating the fkn obvious. Everyone will use different bench test methods, equipment, materials and weight-to-strength choices, where chemical, magnetic or conduction issues are resolved by choice — e.g. carbon fibre over steel. Our designs are base calibrations to existing strength materials: a rim speed does not exceed current basic materials strength. We do not spec methods, and we do not teach welders how to weld.

Section map: §0 this page — §1 the shower law as seated (contents line, the three bodies, the full builder specification re-instructed by the author 29 July 2026 — verbatim) — §2 why pressure beats suction (graded) — §3 the system in the stall — §4 the doctrine record — §6 build sequence — §7 cross-phase — §8 do-not-build — §9 owed — §10 status, provenance, change control.

§1 — THE LAW (AS SEATED, VERBATIM)

THE CONTENTS LINE (verbatim): 'Phase 146: **Zero G Space Shower Systems - Barometric-Flux**. Shapes

the shower stall like a snub-nosed ballistic bullet, placing a high-pressure warm air plenum in the nose and a low-pressure cryogenic rear panel behind an offset colander maze wall. Air and water droplets are naturally drawn along the path of least resistance onto a refrigeration plate to condense silently without suction fans. The rear condenser wall is angled out at one end extending the path of least resistance to a vertical wall gutter engaged by a wall screw guttered rotating pipe, and Archimedes wall screw'

THE MASTHEAD LINE (verbatim): 'Industrial Fluid Engineering: Non-Contact Pressure-Differential Condensation & Aerodynamic Air-Knives [extracted verbatim from the combined masthead line]'

THE CORE OBJECTIVE (verbatim): 'Core Objective: Noise-Free Zero-G Fluid Evacuation via Natural Barometric Pathing [extracted verbatim from the combined masthead line]'

THE SNUB-NOSED BULLET ENCLOSURE GEOMETRY (VERBATIM)

* Rationale: Rejects loud, energy-hogging mechanical vacuum turbines and suction hoses to eliminate surface-tension skin-blanketing water traps and noise pollution across habitation decks. * High-Pressure Semicircular Nose: Positions handheld nozzles and low-speed warm air plenums within a front semicircular dome to establish a localized high-pressure zone. * Low-Pressure Cryogenic Tail: Links the wide rectangular rear panels directly to Phase 96 cold sinks utilizing natural thermal contraction to draw all fluid droplets down the line.

THE PERFORATED COLANDER NON-RETURN MAZE (VERBATIM)

* Offset Panel Fluid Traps: Constructs the rear barrier out of vertical panels spaced 1 inch apart with offset 80mm perforations to form a non-line-of-sight mechanical baffle. * Cryogenic Plate Condensers: Mounts a solid aluminum freezing plate behind the colander panels, forcing instant sub-zero hydrostatic condensation of passing steam and moisture masses. * Convex Floor Drainage Channels: Slants the plates and floors toward localized perimeter wall gutters machined with Phase 64/136 concave tracks to route liquid safely to recycling loops.

THE SLIDING POLE AIR-KNIFE ASSEMBLY (VERBATIM)

* Hands-Free Foot Pedal Controls: Mounts a high-efficiency sliding blade fan to a vertical wall pole, regulated via mechanical foot pedals to allow real-time position tracking. * High-Velocity Vertical Airwaves: Projects a razor-thin, uniform air-knife that strips clinging water off the skin, clearing the nose and mouth lines to prevent surface-tension drowning. * Detachable Handheld Pre-Towels: Enables the crew to detach the air-blade unit for manual body sweeps, clearing 95% of residual surface moisture prior to cabin deck transitions.

BUILDER SPECIFICATION — THE ROOM, RE-INSTRUCTED BY ARCHER (29 JULY 2026) (VERBATIM)

- **Geometry:** sub-nosed bullet with angular keyhole sides (no inward curve at the sides); the rear end is 50% wider than the front width before the curve — the widening slows airflow so droplets settle out.
- **Shower head:** bracket-mounted with hose — clip for hands-free hair, lift for directed washing.
- **Air blade:** vertical slide rail on the facing wall carrying a 10-inch wide blade, parked at adjusted face height, detachable on its own hose for full-body pre-towel drying.
- **Warm air array:** constant soft warm flow from overhead and front outlets — steering force, not blast.
- **Floor:** 5-10 degree slope to the rear wall; electromagnetic entry floor with magnetic-grain-soled bath sandals (304-grade with magnetic grain) — one foot always anchored while the other is washed.
- **Perforated wall:** five 316 stainless panels, ~1 inch apart, 80mm holes with 80mm spaces, each offset — no line of sight for a droplet. The uncut section behind every hole carries a pressed dome (conical profile): any direct-line droplet strikes the dome and scatters sideways instead of rebounding — zero bounce-back by geometry, with the pressing work-hardening the panel as a bonus.
- **Cold panel:** one solid holed-free panel ~100mm behind the last perforated panel, held at ~3C by the central system; semicircular floor cut-out below the perforated panels for drainage.
- **Gutter wall:** rear wall tilted 20 degrees — air in from one end, vertical gutter collecting at the other, narrowing centrifuge-style to a pipe with oscillating double-valve seal: one-way water to reclamation, air circuit unbroken.
- **Fittings (Archer, 29 July):** basic handrails for support while washing feet, and a ship's whirly window mounted as the mirror — a spinning clear disc that centrifugally sheds bouncing spray and steam, so the mirror never fogs. A century-proven marine bridge device, promoted to the bathroom.
- **Design law:** three stacked gradients (volume small-to-large, heat-to-cold, pressure high-to-low) — water and steam can only take the path of least resistance, and the path leads to the cold panel and the gutter. And the room is designed for gravity-drive failure ten years out: it must work in zero-G, so it will.

ORIGIN OF THOUGHT — PHASE 146 (VERBATIM)

Archer, 29 July 2026: "the room works on physics laws not mechanics... the room has a 3 phase high low physics pressure design, size small to large, heat to cold and high low air pressure. the water can only travel in the path of least resistance as with the steam. When i have tried to explain things to people for decades about basic physics i always said, 'It's not rocket science' apparently it was." (Full re-instruction recorded verbatim in the Bathroom Book, Section 3.)

Why it matters, in plain English: every terrestrial bathroom leans on gravity; every space-agency bathroom leans on machinery. This room leans on neither — it is built from three gradients that cost no power worth mentioning and cannot break down, because physics laws don't wear out. The warm soft air, the widening room, and the cold back wall are one conveyor belt made of weather. And the closing philosophy is the discipline under the whole fleet: design for the failure — if the gravity drive dies a decade from home, the crew still showers tomorrow.

PHYSICS NOTE ADDENDUM — PHASE 146 (KIMI AUDIT: THE STACKED GRADIENTS) (VERBATIM)

- **The 3C choice is smarter than sub-zero.** Stall air at shower temperature and high humidity has a dew point far above 3C, so condensation is guaranteed; staying above freezing means no ice fouls the panel, the drains, or the valves. The earlier markup's sub-zero plate would have worked but demanded defrost cycles — 3C deletes them.
- **The keyhole widening is a diffuser.** A duct expanding 50% in cross-section slows the flow and recovers pressure — droplets beyond a few microns can no longer follow the airstream and deposit rearward. Volume gradient, thermal gradient, pressure gradient: three independent mechanisms with one shared destination, which is why the room has no single point of failure. Archer's pressed-dome refinement closes the last gap: a flat impingement plate can reflect a normal-incidence droplet straight back out its entry hole, but a conical surface converts every direct hit into an oblique scatter that sheets down into the drainage cut-out — the same physics as profiled mist-eliminator plates, plus work-hardened stiffness from the pressing.
- **The sandals are Phase 32 boot physics in miniature** — ferromagnetic grain soles on an electromagnetic floor, one foot always anchored. The design-for-failure mandate (zero-G function even if a future gravity drive dies) is the correct robustness discipline and should be read into every crew system.

PHASE 146 — AMENDMENT 1: THE KEYHOLE FLARE & THE ONE-FOOT DRAIN END (5 AUGUST 2026) (VERBATIM)

Live instruction, 5 August 2026 — spoken while reviewing the shower's film render; the corrections are design refinements, seated as an amendment, not just render notes.

[*VERBATIM — first pass*]: "back wall goes to the ceiling, drops go through and the holes are much larger, airbalde on vertical pole on hose so it can be hand held same as the shower head"

[*VERBATIM — second pass*]: "bathroom shorten the empty drain end to about 1 foot they cannot see in anyway, and flare out the room a little more after the standing area like a keyhole shape, maximising high low pressure"

The deeper flare after the standing area is a bigger diffuser: flow deceleration scales with the

expansion ratio, so more keyhole means slower air at the panel bank, a smaller droplet cut-size, and better pressure recovery — 'maximising high low pressure' is the stacked-gradient law getting a wider third step. The one-foot drain end is volume reclaimed: the offset maze already guarantees no line of sight ('they cannot see in anyway'), so the dead section beyond the cold plate only needs its engineering length — plate, channel, service hand — and the rest was empty ship. The full-height back wall closes the last leak path: steam climbs in a warm room, and a panel bank that stops short of the ceiling is a chimney over the maze; to the ceiling, every droplet's only road is through the holes. Larger holes keep the non-return offset geometry while dropping flow resistance — the maze works on offset, not on smallness. And the blade on its pole-and-hose, handheld like the shower head, is the spec's own detachable blade confirmed and simplified: one fitting language for both water and air.

- Spec deltas, seated: rear drain end shortened to ~1 foot beyond the cold plate; room flare after the standing area increased (keyhole profile, wider than the original 50%); perforated panel bank runs full height to the ceiling; perforation diameter seated at the law — 80mm holes with 80mm spacings (big enough for low resistance, offset kept so no sight line); air blade on a vertical pole with hose, handheld, same fitting family as the shower head. Perforation law: 80mm holes with 80mm spaces.
- **Bench item added:** perforation size vs droplet capture — larger holes cut resistance but each hole is a straighter shot; the offset spacing and the widening flare are what keep the guarantee. Run the droplet-tracking numbers once at the new hole size; if capture drops, the answer is more offset, not smaller holes.

§2 — WHY IT WORKS (THE PHYSICS, GRADED)

PRESSURE-DIFFERENTIAL TRANSPORT, AFFIRMED: aimed jets plus steering airflow plus an air-knife blade move water rearward without any suction machinery — Phase 145's momentum directive executed as architecture. The vet's grade stands: the pressure-driven zero-G shower is coherent.

THE SETTLING GEOMETRY, AFFIRMED AS SOUND AERODYNAMICS: the widened rear slows the stream so droplets fall out of it — expansion-chamber settling, the same physics as every industrial knock-out drum.

THE NON-RETURN MAZE, AFFIRMED: offset perforated panels with pressed domes behind each hole give no droplet a line of sight — mechanical baffling, no moving parts; and the 3°C condenser plate is the correct terminal sink: vapor leaves the airstream where the cold surface is.

THE ANCHORED FOOT, AFFIRMED AS THE HUMAN FACTOR: magnetic sandals on an electromagnetic floor, one foot anchored while the other is washed — the stability problem solved at the sole, where it lives. The owed item is sizing, not concept: flow rates, blade duty, and condenser capacity need their numbers — §9.

§3 — THE SYSTEM IN THE STALL

- THE SHELL: snub-nosed bullet with angular keyhole sides — rear half wider, air slowed, droplets settled.
- THE JETS: handheld high-pressure head on a bracket — clip for hair, lift for directed washing.
- THE BLADE: 10-inch air-knife on a vertical rail, foot-pedal positioned, detachable for pre-towel drying.
- THE WARMTH: constant soft warm flow overhead and front — steering force, not blast.
- THE FLOOR: 5-10 degree slope, electromagnetic plate, magnetic-grain sandals — one foot always anchored.
- THE MAZE: five offset 316 perforated panels, pressed domes behind every hole — no line of sight.
- THE COLD: solid panel at $\sim 3^{\circ}\text{C}$ behind the maze — the vapor's terminal sink; gutter wall tilted 20 degrees, draining centrifuge-style.

§4 — THE DOCTRINE RECORD (HOW THE BOOK ALREADY RUNS ON IT)

- THE PRESSURE-NOT-SUCTION DOCTRINE (seated with the phase): turbines and hoses are the noise and energy budget being deleted — the Δp is manufactured at the nose, not borrowed from a pump at the rear.
- THE STEERING-NOT-BLAST DOCTRINE (the warm array): the airflow aims the water; it does not assault the bather.
- THE NO-LINE-OF-SIGHT DOCTRINE (the maze): if a droplet can see through the wall, the wall is wrong — geometry as the filter.
- THE ANCHOR-ONE-FOOT DOCTRINE (the floor): stability is a sole problem — the magnetic sandal is the honest fix, and it is seated in the builder spec.

§5 — VALIDATION: the grade and its full record live in the master and the restart.

§6 — BUILD SEQUENCE

- STEP 1 — Form the shell: bullet geometry with keyhole sides; rear-half widening to the settling spec.
- STEP 2 — Fit the jets and blade: head bracket, rail, pedal linkage; blade duty mapped across positions.
- STEP 3 — Balance the warmth: steering-flow rates set — steering force verified, blast ruled out by gauge.
- STEP 4 — Wire the floor: electromagnetic plate and sandal qualification; one-foot drill rehearsed.

- STEP 5 — Bench the maze and the cold: droplet penetration trials across the panels; condenser capacity versus shower duty — §9's owed sizing.
- STEP 6 — Publish the ledger: shell drawings, flow maps, condensation curves — open under the license.

§7 — CROSS-PHASE (INLINED IN FULL)

- PHASE 145 — the momentum directive (GP-IM-145): the law this stall executes.
- PHASE 137 — the waste core (GP-IM-137): the hygiene envelope's other half.
- PHASE 130 — the recirculation networks (GP-IM-130): where the collected water goes.
- PHASE 136 — the bottle (GP-IM-136): the drinking end of the same water.
- PHASE 131 — the sleep carousel (GP-IM-131): the rest of the crew-care suite.

§8 — DO-NOT-BUILD

- DO NOT suck the shower — turbines and hoses are the deleted noise and energy budget.
- DO NOT blast the bather — the warm array steers; if it pushes, it is wrong.
- DO NOT give a droplet line of sight — the maze and the domes are the spec, not a suggestion.
- DO NOT skip the cold plate's numbers — condenser capacity versus duty is owed; a warm condenser is a fog machine.

§9 — OWED

OWED: the flow and condensation sizing — jet flow rates against air-knife duty against condenser capacity at ~3°C, across the full shower cycle, with gutter drainage rates at the 20-degree tilt. Seats additively when benched. The 12 August naming batch and 15 August kills are carried inline in §1.

§10 — STATUS / PROVENANCE / CHANGE CONTROL

STATUS: CHAPTER GP-IM-146, v1.0 — CURRENT, seated law, master v2.1 (numerical print order). Built 23 August 2026, nineteenth of the 20-chapter 'ok next 20' shift (the author's order, 23 August 2026: 'ok next 20'). First version until publication — 'until i publish it, it is still first version, otherwise is just typing' (his law, 22 August 2026). AMENDED 23 AUGUST 2026, SITTING 586: the author's edit pass on the masthead seated verbatim above (his PDF conversion, 6.14pm) — four edits graded, one flag (perforation diameter: the spec delta governs). Still v1.0 under first-version law — unpublished.

PROVENANCE — THE STANDING ORDERS, VERBATIM: the town-trip order (seated in sitting 519): 'ok i have to go to town, i have converted them to pdfs now and am uploading, do the next 10 phases and i will read them when i get back' — the order that opened the chapter program; the follow-on orders (seated in sittings 537, 557, 561, 562, 564, 566, 572 and 576): 'ok next 10' / 'ok next ten' / 'all good next 10' / '10 more before go to the old lady for dinner' / 'do another 20 while i away so we clear the 100 mark for the day when i get back' / 'ok next 20'.

CHANGE CONTROL: this design changes only on the Author's orders, seated as sittings in the master record. If the master and this chapter ever disagree, THE MASTER WINS. This is a workshop chapter — it carries the current design only; the full change record lives in the master and the restart.

END OF CHAPTER 146